

Curriculum Vita

Vyacheslav Kungurtsev

SUMMARY

Researcher working on Mathematical Programming and Numerical Optimization with applications to Artificial Intelligence, Machine Learning, and Process Control Engineering. More than a decade of experience in formulating novel hardware-aware (HPC, embedded, parallel, distributed, decentralized) algorithms, developing rigorous theoretical convergence and learning guarantees, and testing/validating their performance for understanding and managing scientific systems, with deliberate and principled leverage of both scientific domain knowledge and data.

PERSONAL AND CONTACT INFORMATION:

slava@alumni.duke.edu

Ve Smeckach 16, 11000 Prague, Czechia

(+420) 730-636-921 (Mobile)

Citizen of the United States of America. Permanent Resident in Czechia

EDUCATION

University of California, San Diego

Ph.D. Mathematics with a specialization in computational science, February 2013, GPA 3.95

Duke University

B.S. Mathematics, with Economics minor. May 2007, GPA 3.65

CAREER:

Czech Technical University in Prague,

Faculty of Electrical Engineering Department of Computer Science:

-Tenured Researcher (2025-present)

-Researcher (2019 to 2024)

-Postdoctoral Fellow (2014-2019)

KU Leuven, Department of Electrical Engineering / Department of Computer Science

-Postdoctoral Fellow – (2013 – 2014)

Advisors: Moritz Diehl, Wim Michiels, Stefan Vanderwalle

University of California San Diego,

-Graduate Researcher – Mathematics (2007 – 2013)

Advisor: Philip Gill

SELECTED PUBLICATIONS:

- *The stochastic Ravine accelerated gradient method with general extrapolation coefficients* H Attouch, J Fadili, V Kungurtsev. Journal of Optimization Theory and Applications 209:5, (2026)
- *A stochastic-gradient-based interior-point algorithm for solving smooth bound-constrained optimization problems.* FE Curtis, V Kungurtsev, DP Robinson, Q Wang. SIAM Journal on Optimization 35 (2), 1030-1059 (2025)
- *Towards diverse device heterogeneous federated learning via task arithmetic knowledge integration.* M Morafah, V Kungurtsev, H Chang, C Chen, B Lin. Advances in Neural Information Processing Systems 37, 127834-127877 (2024)
- *Efficient dataset distillation via minimax diffusion.* J Gu, S Vahidian, V Kungurtsev, H Wang, W Jiang, Y You, Y Chen. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (2024)
- *Decentralized bayesian learning with metropolis-adjusted Hamiltonian monte carlo* V Kungurtsev, A Cobb, T Javidi, B Jalaian. Machine Learning 112 (8), 2791-2819 (2023)

- *On the ergodic control of ensembles in the presence of non-linear filters.* V Kungurtsev, J Marecek, R Ghosh, R Shorten. *Automatica* 152, 110946 (2023)
- *Simultaneous online model identification and production optimization using modifier adaptation.* J Matias, V Kungurtsev, M Egan. *Journal of Process Control* 110, 110-120 (2022)
- *A Stochastic Levenberg--Marquardt Method Using Random Models with Complexity Results.* EH Bergou, Y Diouane, V Kungurtsev, CW Royer. *SIAM/ASA Journal on Uncertainty Quantification* 10 (1), 507-536 (2022)
- *Mean-field analysis of piecewise linear solutions for wide ReLU networks.* A Shevchenko, V Kungurtsev, M Mondelli. *Journal of Machine Learning Research* 23 (130), 1-55 (2022)
- *Convergence and complexity analysis of a Levenberg--Marquardt algorithm for inverse problems.* EH Bergou, Y Diouane, V Kungurtsev. *Journal of Optimization Theory and Applications* 185 (3), 927-944 (2020) (110 Citations (Google Scholar))
- *Asynchronous parallel algorithms for nonconvex optimization.* L Cannelli, F Facchinei, V Kungurtsev, G Scutari. *Mathematical Programming* 184 (1), 121-154 (2020) (106 Citations (Google Scholar))

FUNDING:

- PI for CTU on TACR Project for Energy Optimization 2023-2025
- PI for CTU for SoftMaxx-funded project on AI Alignment led by the Institute for Advanced Consciousness Studies in LA, USA 2024-2025
- CINECA Supercomputing Center Project Award: 10000 GPU Hours. 2023
- Individual Collaborative Project: Cisco-CTU Work Program 5 Industrial Grant 2015

SOFTWARE

Joint with PhD student Anshu Kumar: A C++23 library for a modular approach to solving different computational fluid dynamics PDEs with different numerical methods.

<https://github.com/Anshu8879065/FlexCFD>

Supervising (ongoing)

<https://github.com/cheropnai/hospital-robot-system> Planning-Control-RL simulation of Robotic Health Care (supervising Monicah Cheronoi)

<https://github.com/rxymitchy/prospect-theory/> Repeated noncooperative games with AI Agents with Expected Utility preferences and Humans with Prospect Theoretic Preferences (supervising Dylan Waldner and Michelle Luciana)

Past

Contributed a nonlinear optimization solver in C++ to CasADi in 2013

SUPERVISION

Doctoral Students

Graduated

-Co-advised with Jakub Marecek (CTU and CEZ): Antonio Bellon *Parametric Semidefinite Programming* Czech Technical University. 2019-2024

Current

-Two as primary supervisor: Anshu Kumar (starting 2025) Harsh Choudhary (starting 2023) *Machine Learning for Scientific Domains*

- Fadwa Idlahcen. Co-advising for Ondrej Kuzelka (CTU) (starting 2023). *Probabilistic Graphical Models and Machine Learning for Hybrid Domains*

-Azad Afandizada. Co-advising for Andre Sopczak (CTU and CERN): starting 2025. *Machine Learning for Accelerating Leptoquark Discovery*

Masters Students

Current

-Alessandro Di Frenna “Hybrid RL-MPC for Autonomous Embodied AI: Integrating Vision-Language-Action (VLA) models with Hybrid Reinforcement Learning-Model Predictive Control (RL-MPC) to improve decision-making in Autonomous embodied systems” University of Padua 2026

Graduated

-Alberto Schiabel “Advances in Submodularity: Optimization and Probabilistic Models” Masters Thesis, University of Padua 2022

-Jacopo Guidolin “Jump Langevin for Multimodal Bayesian Sampling” Masters Thesis. University of Padua 2023.

-A. Van Barel “Robust Optimization of Systems Described by Stochastic Partial Differential Equations” Masters Thesis, KULeuven 2015

Other

Visiting Scholar: Associate Professor Songqian Qiu from the University of Mining and Technology, China 2022- 2023. *Optimization with Stochastic Objective Functions and Deterministic System Constraints*

TALKS IN CONFERENCES

Invited Talks

-2024: International Symposium on Mathematical Programming. Riemannian Stochastic Approximation of Tame Functions

-2023: EUROPT and the SIAM Optimization Conference. Retraction Based Direct Search Methods for Derivative Free Riemannian Optimization

-2022: International Conference for Continuous Optimization (ICCOPT) Scaling the Wild: Decentralizing Hogwild!-style Shared-memory SGD

-2021: INFORMS Optimization Society. Local SGD Meets Asynchrony

-2021: SIAM Optimization Conference. Zero Order Stochastic Weakly Convex Optimization

-2020: Games, Dynamics and Optimization. A Stochastic Line Search Algorithm with Second Order Results

-2019: ICCOPT. A Stochastic Line Search Algorithm with Second Order Results

-2014 European Conference on Control and Optimization. Parametric Nonlinear Programming and Model Predictive Control

-2013 ICCOPT. Local convergence of a primal-dual augmented Lagrangian algorithm with a stabilized SQP subproblem

Contributed Talks

-2024: (Conference Publication) International Conference of Distributed Computing Systems. Federated SGD with Local Asynchrony

-2021: (Conference Publication) AAAI. Asynchronous Optimization Methods for Efficient Training of Deep Neural Networks with Guarantees

-2020: (Conference Publication) CDC. Two Step Preprocessing for Semidefinite Programming

-2017: (Conference Publication) IFAC World Congress. Spectral Optimization for Design of Input Shapers

-2014: (Conference Publication) ECC. Linear Convergence of Adjoint-Based Distributed Multiple Shooting

-2013: Erice Workshop on Optimization. An Exact Hessian SQP Method Superlinearly Convergent to Second Order Points

Short Term Positions:

-Argonne National Lab, Research Aide, Argonne, IL (June-September 2010)
-Microsoft Research Asia, Project Manager, Beijing, China (June-August 2007)
-UCLA/Lawrence Livermore National Lab, Research Intern, Los Angeles/Livermore, CA (June-August 2006)

Complete Publication List

(imported from Google Scholar)

Journal Articles

Publication	Cited	Year
The stochastic Ravine accelerated gradient method with general extrapolation coefficients H Attouch, J Fadili, V Kungurtsev. Journal of Optimization Theory and Applications	2	2026
Parallel variational quantum algorithms with gradient-informed restart to speed up optimization in the presence of barren plateaus D Mastropietro, G Korpas, V Kungurtsev, J Marecek. IEEE Transactions on Quantum Engineering	22	2026
Antiproliferative activity, phytochemistry, network pharmacology, molecular docking and gene expression analysis of Maerua edulis extracts against human cervical cancer cell line. Inyani John Lino Lagu, Rakita Letoluo, Sally Wambui Kamau, James Menni Kuria, Fredrick Mutie Musila, Vyacheslav Kungurtsev, Dorothy W. Nyamai and Sospeter Ngoci Njeru. BMC Complementary Medicine and Therapies.		2026
Probabilistic iterative hard thresholding for sparse learning M Bergamaschi, A Cristofari, V Kungurtsev, F Rinaldi Computational Optimization and Applications, 1-27		2025
Leveraging Gradient Noise for Detection and Filtering of Byzantine Clients L Errami, V Kungurtsev, EH Bergou. IEEE Access		2025
Optimal Control of Two-Phase Membrane Problem F Bozorgnia, V Kungurtsev Applied Mathematics & Optimization 92 (1), 1-23		2025
A stochastic-gradient-based interior-point algorithm for solving smooth bound-constrained optimization problems FE Curtis, V Kungurtsev, DP Robinson, Q Wang SIAM Journal on Optimization 35 (2), 1030-1059	15	2025
Truss topology design under harmonic loads: peak power minimization with semidefinite programming S Ma, J Mareček, V Kungurtsev, M Tyburec Structural and Multidisciplinary Optimization 68 (2), 40		2025
Parametric semidefinite programming: geometry of the trajectory of solutions A Bellon, D Henrion, V Kungurtsev, J Mareček Mathematics of Operations Research 50 (1), 410-430	4	2025
Retraction-Based Direct Search Methods for Derivative Free Riemannian Optimization V Kungurtsev, F Rinaldi, D Zeffiro Journal of Optimization Theory and Applications 203 (2), 1710-1735	3	2024
Iteration complexity of variational quantum algorithms V Kungurtsev, G Korpas, J Marecek, EY Zhu Quantum 8, 1495	19	2024
Mini-batch stochastic subgradient for functional constrained optimization NK Singh, I Necoara, V Kungurtsev Optimization 73 (7), 2159-2185	8	2024
Inexact direct-search methods for bilevel optimization problems Y Diouane, V Kungurtsev, F Rinaldi, D Zeffiro Computational Optimization and Applications 88 (2), 469-490	1	2024
Trilevel and multilevel optimization using monotone operator theory A Shafiei, V Kungurtsev, J Marecek Mathematical Methods of Operations Research 99 (1), 77-114	9	2024
Time-varying semidefinite programming: Path following a burer–monteiro factorization	7	2024

A Bellon, M Dressler, V Kungurtsev, J Mareček, A Uschmajew SIAM Journal on Optimization 34 (1), 1-26		
Pathfollowing for parametric mathematical programs with complementarity constraints V Kungurtsev, J Jäschke Optimization and Engineering 24 (4), 2795-2826		2023
Decentralized bayesian learning with metropolis-adjusted hamiltonian monte carlo V Kungurtsev, A Cobb, T Javidi, B Jalaian Machine Learning 112 (8), 2791-2819	9	2023
Regularized quasi-monotone method for stochastic optimization V Kungurtsev, V Shikhman Optimization Letters 17 (5), 1215-1228	1	2023
On the ergodic control of ensembles in the presence of non-linear filters V Kungurtsev, J Marecek, R Ghosh, R Shorten Automatica 152, 110946	16	2023
Decentralized asynchronous nonconvex stochastic optimization on directed graphs V Kungurtsev, M Morafah, T Javidi, G Scutari IEEE Transactions on Control of Network Systems 10 (4), 1796-1804	15	2023
Distributed stochastic nonsmooth nonconvex optimization V Kungurtsev Operations Research Letters 50 (6), 627-631	3	2022
Mean-field analysis for heavy ball methods: Dropout-stability, connectivity, and global convergence D Wu, V Kungurtsev, M Mondelli Transactions in Machine Learning Research	2	2022
A subsampling line-search method with second-order results EH Bergou, Y Diouane, V Kunc, V Kungurtsev, CW Royer INFORMS Journal on Optimization 4 (4), 403-425	32	2022
Diminishing stepsize methods for nonconvex composite problems via ghost penalties: from the general to the convex regular constrained case F Facchinei, V Kungurtsev, L Lampariello, G Scutari Optimization Methods and Software 37 (4), 1242-1268	3	2022
Simultaneous online model identification and production optimization using modifier adaptation J Matias, V Kungurtsev, M Egan Journal of Process Control 110, 110-120	4	2022
A Stochastic Levenberg--Marquardt Method Using Random Models with Complexity Results EH Bergou, Y Diouane, V Kungurtsev, CW Royer SIAM/ASA Journal on Uncertainty Quantification 10 (1), 507-536	18	2022
Mean-field analysis of piecewise linear solutions for wide ReLU networks A Shevchenko, V Kungurtsev, M Mondelli Journal of Machine Learning Research 23 (130), 1-55	23	2022
A zeroth order method for stochastic weakly convex optimization V Kungurtsev, F Rinaldi Computational Optimization and Applications 80 (3), 731-753	9	2021
Complexity iteration analysis for strongly convex multi-objective optimization using a Newton path-following procedure EH Bergou, Y Diouane, V Kungurtsev Optimization Letters 15 (4), 1215-1227	3	2021
Ghost penalties in nonconvex constrained optimization: Diminishing stepsizes and iteration complexity F Facchinei, V Kungurtsev, L Lampariello, G Scutari Mathematics of Operations Research 46 (2), 595-627	41	2021
A nonmonotone matrix-free algorithm for nonlinear equality-constrained least-squares problems EH Bergou, Y Diouane, V Kungurtsev, CW Royer SIAM Journal on Scientific Computing 43 (5), S743-S766	2	2021
Convergence rate for diminishing stepsize methods in nonconvex constrained optimization via ghost penalties F Facchinei, V Kungurtsev, L Lampariello, G Scutari Atti della Accademia Peloritana dei Pericolanti-Classe di Scienze Fisiche ...	1	2020
Asynchronous parallel algorithms for nonconvex optimization L Cannelli, F Facchinei, V Kungurtsev, G Scutari	106*	2020

Mathematical Programming 184 (1), 121-154		
Asynchronous optimization over graphs: Linear convergence under error bound conditions	26	2020
L Cannelli, F Facchinei, G Scutari, V Kungurtsev		
IEEE Transactions on Automatic Control 66 (10), 4604-4619		
Convergence and complexity analysis of a Levenberg–Marquardt algorithm for inverse problems	98	2020
EH Bergou, Y Diouane, V Kungurtsev		
Journal of Optimization Theory and Applications 185 (3), 927-944		
Fast sensitivity-based economic model predictive control for degenerate systems	3	2020
E Suwartadi, V Kungurtsev, J Jäschke		
Journal of Process Control 88, 54-62		
Iteration complexity of a fixed-stepsize SQP method for nonconvex optimization with convex constraints	4	2020
F Facchinei, V Kungurtsev, L Lampariello, G Scutari		
Numerical analysis and optimization, 109-120		
Second-order guarantees of distributed gradient algorithms	72	2020
A Daneshmand, G Scutari, V Kungurtsev		
SIAM Journal on Optimization 30 (4), 3029-3068		
A shifted primal-dual penalty-barrier method for nonlinear optimization	25	2020
PE Gill, V Kungurtsev, DP Robinson		
SIAM Journal on Optimization 30 (2), 1067-1093		
A stabilized SQP method: superlinear convergence	49	2017
PE Gill, V Kungurtsev, DP Robinson		
Mathematical Programming 163 (1), 369-410		
Sensitivity-based economic NMPC with a path-following approach	24	2017
E Suwartadi, V Kungurtsev, J Jäschke		
Processes 5 (1), 8		
A predictor-corrector path-following algorithm for dual-degenerate parametric optimization problems	33	2017
V Kungurtsev, J Jäschke		
SIAM Journal on Optimization 27 (1), 538-564		
A stabilized SQP method: global convergence	58	2017
PE Gill, V Kungurtsev, DP Robinson		
IMA Journal of Numerical Analysis 37 (1), 407-443		
Hybrid random/deterministic parallel algorithms for convex and nonconvex big data optimization	84	2015
A Daneshmand, F Facchinei, V Kungurtsev, G Scutari		
IEEE Transactions on Signal Processing 63 (15), 3914-3929		
Sequential quadratic programming methods for parametric nonlinear optimization	38*	2014
V Kungurtsev, M Diehl		
Computational Optimization and Applications 59 (3), 475-509		
Numerical bifurcation analysis of conformal formulations of the Einstein constraints	4	2011
M Holst, V Kungurtsev		
Physical Review D—Particles, Fields, Gravitation, and Cosmology 84 (12), 124038		

Other Publications – Conference Papers and Book Chapters

Publication	Cited	Year
CONCORD: Concept-Informed Diffusion for Dataset Distillation		
J Gu, H Wang, R Jia, S Vahidian, V Kungurtsev, W Jiang, Y Chen	1	2026
Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision		
Towards diverse device heterogeneous federated learning via task arithmetic knowledge integration	8	2024
M Morafah, V Kungurtsev, H Chang, C Chen, B Lin		
Advances in Neural Information Processing Systems 37, 127834-127877		
Unlocking the potential of federated learning: The symphony of dataset distillation via deep generative latents	12	2024
Y Jia, S Vahidian, J Sun, J Zhang, V Kungurtsev, NZ Gong, Y Chen		
European Conference on Computer Vision, 18-33		
Federated SGD with Local Asynchrony	3*	2024
B Chatterjee, V Kungurtsev, D Alistarh		

2024 IEEE 44th International Conference on Distributed Computing Systems ...		
Efficient dataset distillation via minimax diffusion		
J Gu, S Vahidian, V Kungurtsev, H Wang, W Jiang, Y You, Y Chen	56	2024
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern ...		
Efficient distribution similarity identification in clustered federated learning via principal angles between client data subspaces		
S Vahidian, M Morafah, W Wang, V Kungurtsev, C Chen, M Shah, B Lin	93	2023
Proceedings of the AAAI conference on artificial intelligence 37 (8), 10043 ...		
When do curricula work in federated learning?		
S Vahidian, S Kadaveru, W Baek, W Wang, V Kungurtsev, C Chen, ...	13	2023
Proceedings of the IEEE/CVF International Conference on Computer Vision ...		
A sensitivity assisted alternating directions method of multipliers for distributed optimization		
D Krishnamoorthy, V Kungurtsev	5*	2022
2022 IEEE 61st Conference on Decision and Control (CDC), 295-300		
Asynchronous optimization methods for efficient training of deep neural networks with guarantees		
V Kungurtsev, M Egan, B Chatterjee, D Alistarh	8*	2021
Proceedings of the AAAI Conference on Artificial Intelligence 35 (9), 8209-8216		
Elastic consistency: A practical consistency model for distributed stochastic gradient descent		
G Nadiradze, I Markov, B Chatterjee, V Kungurtsev, D Alistarh	28*	2021
Proceedings of the AAAI Conference on Artificial Intelligence 35 (10), 9037-9045		
A two-step pre-processing for semidefinite programming		
V Kungurtsev, J Marecek	11	2020
2020 59th IEEE Conference on Decision and Control (CDC), 384-389		
Lifted weight learning of markov logic networks (revisited one more time)		
O Kuzelka, V Kungurtsev, Y Wang	8	2020
International Conference on Probabilistic Graphical Models, 269-280		
Reinforcement learning based on real-time iteration NMPC		
M Zanon, V Kungurtsev, S Gros	20	2020
ifac-Papersonline 53 (2), 5213-5218		
Lifted weight learning of markov logic networks revisited		
O Kuzelka, V Kungurtsev	8	2019
The 22nd International Conference on Artificial Intelligence and Statistics ...		
Second-order guarantees of gradient algorithms over networks		
A Daneshmand, G Scutari, V Kungurtsev	17	2018
2018 56th Annual Allerton Conference on Communication, Control, and ...		
Input shaper optimization with a constraint on the spectrum distribution		
V Kungurtsev, D Pilbauer, T Vyhřídál, M Hromčík, W Michiels	4	2017
IFAC-PapersOnLine 50 (1), 13324-13329		
Capacity sensitivity in additive non-gaussian noise channels		
M Egan, SM Perlaza, V Kungurtsev	17*	2017
2017 IEEE International Symposium on Information Theory (ISIT), 416-420		
An SL/QP Algorithm for Minimizing the Spectral Abscissa of Time Delay Systems		
V Kungurtsev, W Michiels, M Diehl		2017
Time Delay Systems: Theory, Numerics, Applications, and Experiments, 33-45		
Asynchronous parallel nonconvex large-scale optimization		
L Cannelli, F Facchinei, V Kungurtsev, G Scutari	8	2017
2017 IEEE International Conference on Acoustics, Speech and Signal ...		
Parallel asynchronous lock-free algorithms for nonconvex big-data optimization		
L Cannelli, G Scutari, F Facchinei, V Kungurtsev	6	2016
2016 50th Asilomar Conference on Signals, Systems and Computers, 1009-1013		
Flexible selective parallel algorithms for big data optimization		
A Daneshmand, F Facchinei, V Kungurtsev, G Scutari	8	2014
2014 48th Asilomar conference on signals, systems and computers, 3-7		
Linear convergence of distributed multiple shooting		
V Kungurtsev, A Kozma, M Diehl		2014
2014 European Control Conference (ECC), 2874-2879		
A regularized SQP method with convergence to second-order optimal points	13	2013
PE Gill, V Kungurtsev, DP Robinson		

[Second-derivative sequential quadratic programming methods for nonlinear optimization](#)

V Kungurtsev [18](#) 2013
University of California, San Diego

[Disambiguation of the DBLP Database](#)

M Alvisio, C Nathe, C Kim, V Kungurtsev [1](#) 2009

Preprints (mostly under review) and Patent Applications

Publication Cited Year

[Numerical Algorithms for Partially Segregated Elliptic Systems](#)

F Bozorgnia, A Arakelyan, V Kungurtsev, J Valdman 2026
arXiv preprint arXiv:2603.05991

[Multistage Stochastic Programming for Rare Event Risk Mitigation in Power Systems Management](#). D

Mastropietro, V Kungurtsev. arXiv preprint arXiv:2603.04734 2026

[Towards Tsallis Fully Probabilistic Design](#)

V Kungurtsev, G Russo. arXiv preprint arXiv:2602.23892 2026

[An efficient penalty decomposition algorithm for minimization over sparse symmetric sets](#). A Mousavi,

M Kimiaei, S Babaie-Kafaki, V Kungurtsev. arXiv preprint arXiv:2601.12383 2026

[A Three-Tier Time-Scale Architecture for Controlling Complex Nonlinear Systems](#)

V Kungurtsev. arXiv preprint arXiv:2601.01621 2026

[Survey of Mathematical Models of and Numerical Methods for Fluid Dynamics Water Engineering](#) A

Kumar, K Olimba, V Kungurtsev, FV Difonzo. arXiv preprint arXiv:2512.16351 2025

[Towards Real Time Control of Water Engineering with Nonlinear Hyperbolic Partial Differential](#)

[Equations](#) F DiFonzo, M Holst, M Kimiaei, V Kungurtsev, S Qiu 2025
arXiv preprint arXiv:2512.14387

[Fractional Heat Kernel for Semi-Supervised Graph Learning with Small Training Sample Size](#) F

Bozorgnia, V Kungurtsev, S Kadyrov, M Yousefnezhad. arXiv preprint arXiv:2510.04440 2025

[Machine Learning Algorithms for Improving Black Box Optimization Solvers](#)

M Kimiaei, V Kungurtsev. arXiv preprint arXiv:2509.25592 2025

[Quasi-Ergodic Control of Multi-Periodic Autoregressive Processes: Formulation and Examples](#) V

Kungurtsev arXiv preprint arXiv:2509.24729 2026

[Machine Learning Algorithms for Improving Exact Classical Solvers in Mixed Integer Continuous](#)

[Optimization](#) M Kimiaei, V Kungurtsev, B Olimba 2025
arXiv preprint arXiv:2508.06906

[Binarizing Physics-Inspired GNNs for Combinatorial Optimization](#)

M Krutský, G Šír, V Kungurtsev, G Korpas 2025
arXiv preprint arXiv:2507.13703

[A Survey of Multi Agent Reinforcement Learning: Federated Learning and Cooperative and Noncooperative Decentralized Regimes](#)

K Cheruiyot, N Kiprotich, V Kungurtsev, K Mugo, V Mwirigi, M Ngesa [2](#) 2025
arXiv preprint arXiv:2507.06278

[Mission-Aligned Learning-Informed Control of Autonomous Systems: Formulation and Foundations](#)

V Kungurtsev, A Anand, S Gros, H Tian, H Hamedmoghadam 2025
arXiv preprint arXiv:2507.04356

[AI Alignment Foundations from First Principles: AI Ethics, Human and Social Considerations](#)

V Kungurtsev [1](#) 2025
<https://philpapers.org/archive/KUNAAF.pdf>

[Optimisation techniques for variational quantum algorithms using noisy quantum processing hardware](#)

G KORPAS, J Marecek, P INTALLURA, D MASTROPIETRO, ... 2025
US Patent App. 18/963,276

[CONCORD: Concept-Informed Diffusion for Dataset Distillation](#)

J Gu, H Wang, R Jia, S Vahidian, V Kungurtsev, W Jiang, Y Chen 2025
arXiv preprint arXiv:2505.18358

[Systems and methods for federated computer processing architecture and coordinated broadcast data messaging](#)

G KORPAS, J KULCSAR, V KUNGURTSEV, W SHOOSMITH 2025
US Patent App. 19/007,112

Direct-search methods for decentralized blackbox optimization		
EH Bergou, Y Diouane, V Kungurtsev, CW Royer arXiv preprint arXiv:2504.04269		2025
Undecidable problems associated with variational quantum algorithms		
G Korpas, V Kungurtsev, J Mareček arXiv preprint arXiv:2503.23723		2025
Federated Sinkhorn		
J Kulcsar, V Kungurtsev, G Korpas, G Giaconi, W Shoosmith arXiv preprint arXiv:2502.07021		2025
Planting 10 Million Trees with Lukenya University in Kenya: Methodology and Preliminary Observations and Forecasts		
P Kyalo, V Kungurtsev, E Muli, S Pietsch, K Mugo, M Kaigongi, M Muasa, ... bioRxiv, 2025.01. 09.632148		2025
A class of diagonal quasi-Newton penalty decomposition algorithms for sparse bound-constrained nonconvex optimization		
A Mousavi, M Kimiaei, S Babaie-Kafaki, V Kungurtsev		2025
Restarting nonlinear conjugate gradient methods		
M Kimiaei, S Babaie-Kafaki, V Kungurtsev, M Yousefi		2025
Effect of Weather and Soil Management on Medicinal Compound Concentration in East African Arid Acacias: A Simulation Study		
EW Kinyua, S Pietsch, E Muli, V Kungurtsev, S Neru, K Mugo bioRxiv, 2025.01. 10.632480		2025
Cause" is Mechanistic Narrative within Scientific Domains: An Ordinary Language Philosophical Critique of" Causal Machine Learning	2	2025
V Kungurtsev, LC Moore, M Krutsky arXiv preprint arXiv:2501.05844		
AI Alignment Foundation from First Principles: Cybernetic, Mechatronic and Engineering Aspects		
Kungurtsev, Vyacheslav, G Sir, M Krutsky PhilArchive		2025
Learning Generalized Hamiltonians using fully Symplectic Mappings		
H Choudhary, C Gupta, M Leok, G Korpas arXiv preprint arXiv:2409.11138		2024
Dataset distillation from first principles: Integrating core information extraction and purposeful learning	4	
V Kungurtsev, Y Peng, J Gu, S Vahidian, A Quinn, F Idlahcen, Y Chen arXiv preprint arXiv:2409.01410		2024
Empirical Bayes for Dynamic Bayesian Networks Using Generalized Variational Inference		
V Kungurtsev, A Khandelwal, PS Rastogi, B Chatterjee, J Mareček arXiv preprint arXiv:2406.17831		2024
Learning Dynamic Bayesian Networks from Data: Foundations, First Principles and Numerical Comparisons	2	
V Kungurtsev, F Idlahcen, P Rysavy, P Rytir, A Wodecki arXiv preprint arXiv:2406.17585		2024
Robust Quantum Gate Complexity: Foundations	1	
J Aspmann, V Kungurtsev, J Marecek arXiv preprint arXiv:2404.15828		2024
Spectral methods for quantum optimal control: Artificial boundary conditions	3	
A Wodecki, J Marecek, V Kungurtsev, P Eichler, G Korpas, P Intallura arXiv preprint arXiv:2403.14436		2024
Group distributionally robust dataset distillation with risk minimization	9	
S Vahidian, M Wang, J Gu, V Kungurtsev, W Jiang, Y Chen arXiv preprint arXiv:2402.04676		2024
The Effects of Transmission-Rights Pricing on Multi-Stage Electricity Markets		
EB de Saint-Lienard, J Marecek, V Kungurtsev arXiv preprint arXiv:2401.15772		2024
Scheduling a Multi-Product Pipeline: A Discretized MILP Formulation		
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Fleming-viot helps speed up variational quantum algorithms in the presence of barren plateaus	5	2023

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